CS 310 : Automata Theory

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4. Start symbol One of the nonterminals is the start symbol.
   Traditionally S is the start symbol

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- N is the set of nonterminals
- T is the set of terminals,
- P is the set of production rules, and
- $S \in N$ is the start symbol.

Examples of CFGs

Give context-free grammars for the following

1. $L_1 = \{0^n 1^n \mid n \geq 0\}$
2. $L_2 = \{0^i 1^j \mid i < j\}$
3. $L_3 = \{0^i 1^j \mid i > j\}$
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If grammar is obvious from the context we may only write $\Rightarrow$.

Multistep derivations

We define application of $\overset{G}{\Longrightarrow}$ zero or more times as follows.

Definition

$\alpha \overset{G*}{\Longrightarrow} \beta$ if there is a sequence of words $\gamma_1, \ldots \gamma_n$ such that

- $\alpha = \gamma_1$,
- $\beta = \gamma_n$, and
- $\gamma_{i-1} \overset{G}{\Longrightarrow} \gamma_i$ for each $i \in 2 \ldots n$.

Language of a grammar

Definition

Let $G = (N, T, P, S)$ be a CFG. The language of G, denoted $L(G)$, is the set of words (with only terminals!) that have derivations from the start symbol.

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Theorem

$L = L(G)$ for some context-free grammar G iff L is accepted by some NPDA.

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- Programming languages are specified with CFGs : a grammar is often the reference people use to learn a language's syntax.
- Designers of compilers and interpreters typically start by writing a context-free grammar that specifies the language's syntax.
- Most compilers/interpreters contain a parser, which extracts the meaning of a program before generating compiled code or performing interpreted execution.
- A number of methodologies let us construct a parser automatically, once a context-free grammar for the language is available.

More interesting context-free grammar

Example

Consider the language of arithmetic expressions over binary numbers, *e. g.*,

- $10 + 1$
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Formally, the grammar for arithmetic expressions is

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where, P contains the following production rules.

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We can limit the derivation moves and have predictable sequence of derivations, i.e., making derivations deterministic.

Leftmost derivation

Always expand leftmost nonterminal. We denote such derivation by $\xRightarrow{lm}$

Consider again the following arithmetic expression grammar.

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Let us see the leftmost derivation of $10 \times 1 + 11$ in the above grammar.

$$E \overset{lm}{\Longrightarrow} E + E \overset{lm}{\Longrightarrow} E \times E + E \overset{lm}{\Longrightarrow} B \times E + E \overset{lm}{\Longrightarrow} B0 \times E + E \overset{lm}{\Longrightarrow} 10 \times E + E$$

$$\overset{lm}{\Longrightarrow} 10 \times B + E$$

Leftmost derivation

Always expand leftmost nonterminal. We denote such derivation by $\xRightarrow{lm}$

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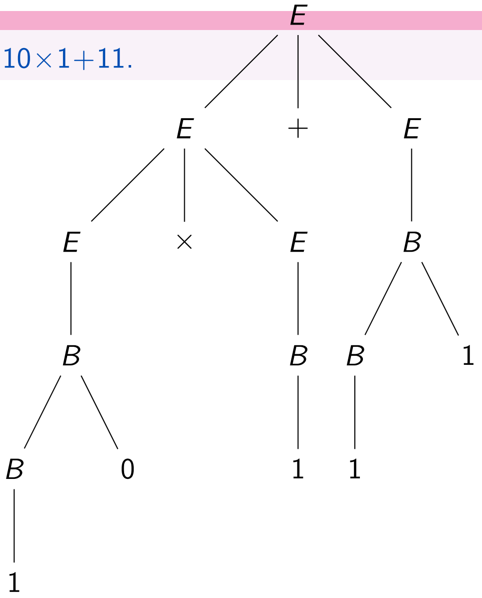
To understand the above intuition, we view the words in derivations as tree.

Example : parse tree

Parse tree which derives $10 \times 1 + 11$.



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Definition

The yield of a parse tree is the word formed by all the leaves from left to right

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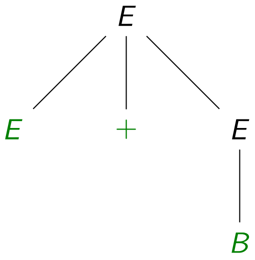
The yield of the following parse tree is $E + B$



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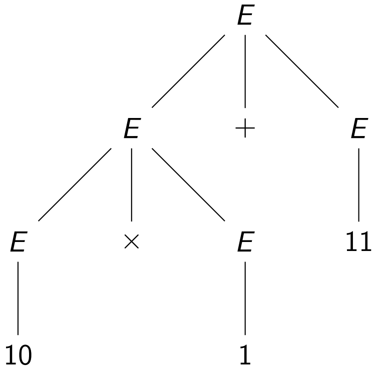
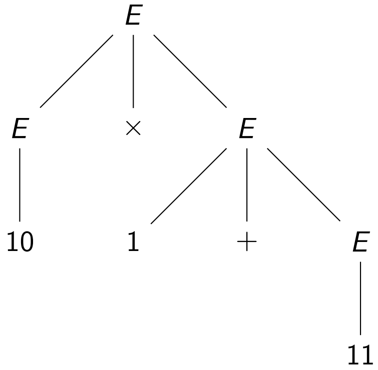


Multiply first



Add first

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$10 \times 1 + 11$ can have the following two interpretations

- 101 (binary 5) — $(10 \times 1) + 11$: multiply first
- 1000 (binary 8) — $10 \times (1 + 11)$: add first

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Remove it for arithmetic expressions, by defining a different grammar!